

AI IN CYBER SECURITY LAB

LAB ASSIGNMENT - 01

End-to-End Mini Project: Building an AI-Driven Cyber Threat Awareness and Detection Prototype

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Assignment Type: Individual

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Academic mini-project demonstrating foundational AI, machine learning, threat analysis, visualization, model evaluation, and ethical considerations in cyber security.

Declaration and Abstract

Student Declaration

I, Bhaumik Senwal, Roll No. 2301730328, submit this report for the AI in Cyber Security Lab (ENSP355). The implementation described here is an educational prototype created for the assignment requirements. The dataset is synthetic and does not contain real credentials, real IP addresses, or personally identifiable information.

Abstract

This mini-project presents an end-to-end prototype for detecting suspicious login events using foundational artificial intelligence and machine learning techniques. The workflow covers dataset preparation, exploratory data analysis, preprocessing, supervised classification, a neural-network prototype, evaluation, threshold analysis, and ethical reflection.

A synthetic dataset containing 6,000 security events was constructed with access hour, failed login attempts, session duration, inbound traffic volume, country, device type, VPN usage, new-device status, and an IP-reputation score. Missing values were deliberately introduced to demonstrate preprocessing. Exploratory visualizations were used to examine class distribution and security-related patterns.

Logistic Regression was implemented as the baseline model, while an MLPClassifier with two hidden layers was used as the neural-network prototype. Because suspicious events represented a minority class, the training set was balanced without altering the untouched test set. Accuracy, precision, recall, and F1-score were used for evaluation.

The project concludes that AI-based cyber security detection should be treated as decision support rather than an unquestionable automated verdict. False alarms, missed threats, biased data, privacy exposure, and limited explainability all require attention before a prototype could be used in a real organization.

Keywords: Artificial Intelligence, Cyber Security, Machine Learning, Logistic Regression, Neural Network, Threat Classification, EDA, Ethical AI

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The assignment requires repository organization, EDA, a supervised ML classifier, a neural-network prototype, performance evaluation, ethical analysis, and a technical PDF report. This report documents each component and is accompanied by the source code, dataset, plots, README, requirements file, and project archive.

1. Introduction and Problem Context

Modern organizations depend on digital infrastructure for authentication, communication, storage, financial activity, and business operations. This creates a large volume of security-relevant events. Login attempts, device changes, and network activity can contain signals associated with suspicious behavior.

The assignment describes threats including phishing, malware, insider misuse, and automated attacks and notes that static rules and signatures can be insufficient against intelligent or adaptive threats. It introduces AI as a way to learn patterns from data, identify abnormal behavior, and assist security teams.

This mini-project simulates the role of an entry-level cyber security analyst who must construct a basic AI-enabled analysis and detection workflow. The emphasis is on understanding the complete pipeline: data, preprocessing, analysis, model, evaluation, and interpretation.

The prototype focuses on login and security events. Potential indicators include repeated failed attempts, unusual access times, use of a new device, VPN usage, and IP reputation. These indicators are not treated as proof of malicious behavior; they are model inputs that may contribute to a risk prediction.

Project research question: Can a simple AI-based classification pipeline identify patterns associated with suspicious login events while demonstrating the practical trade-offs, limitations, and ethical issues involved in automated cyber security decisions?

Expected outcome: a reproducible educational prototype that satisfies the assignment requirements and demonstrates the fundamentals of AI-driven cyber threat analysis.

2. Objectives, Scope and Environment

The assignment learning objectives include implementing AI-driven cyber security analysis pipelines, applying foundational ML concepts, visualizing threat patterns, understanding AI-based detection, and reflecting on ethical and operational challenges.

Project objectives

1. Create a reproducible security-event dataset.
2. Perform EDA and visualize potential threat indicators.
3. Handle missing values, categorical variables, and numerical scaling.
4. Train Logistic Regression as a baseline classifier.
5. Train a neural-network prototype using MLPClassifier.
6. Measure accuracy, precision, recall, and F1-score.
7. Examine threshold effects on false positives and false negatives.
8. Document privacy, bias, explainability, and operational risks.

Scope

This is an educational prototype. It does not perform live packet capture, vulnerability scanning, credential testing, malware execution, or interaction with real systems. The model operates only on a synthetic CSV dataset.

Software environment

Python is used with NumPy, pandas, scikit-learn, matplotlib, Jupyter, and ReportLab. The project is designed to run locally with a standard Python environment.

Repository structure

```
data/cyber_login_events.csv
src/train_models.py
outputs/class_distribution.png
outputs/suspicious_by_hour.png
outputs/failed_attempts_boxplot.png
outputs/confusion_matrix.png
outputs/threshold_analysis.csv
outputs/results.json
README.md
requirements.txt
.gitignore
```

3. Dataset Description

Feature	Type	Description
hour	Numeric	Hour of day (0-23).
failed_attempts	Numeric	Number of failed authentication attempts.
session_duration_min	Numeric	Approximate session duration in minutes.
bytes_in	Numeric	Approximate inbound data volume.
country	Categorical	Synthetic country/category.
device	Categorical	Windows, Linux, macOS, Android or iOS.
vpn_used	Binary	VPN indicator.
new_device	Binary	Whether the device is new.
ip_reputation	Numeric	Synthetic reputation score from 0 to 100.
suspicious	Target	0 = normal, 1 = suspicious.

Dataset size: 6,000 synthetic records.

Missing values: Approximately 2 percent missing values were introduced in selected fields to demonstrate imputation. This is intentional.

Target construction: The synthetic target was generated from combinations of security-relevant indicators. This makes the data useful for supervised-learning demonstrations but means that results must not be interpreted as evidence about real-world attacker behavior.

Privacy: No real usernames, passwords, email addresses, IP addresses, or direct personal identifiers are included.

Data-quality limitations: Real cyber datasets may contain class imbalance, noisy labels, temporal dependence, changing attacks, duplicates, and organizational context. The current dataset captures only a simplified subset.

4. Exploratory Data Analysis

The assignment requires identification of common threat indicators, visualization of suspicious activity, comparison of normal and suspicious behavior, and plots of attack-related features.

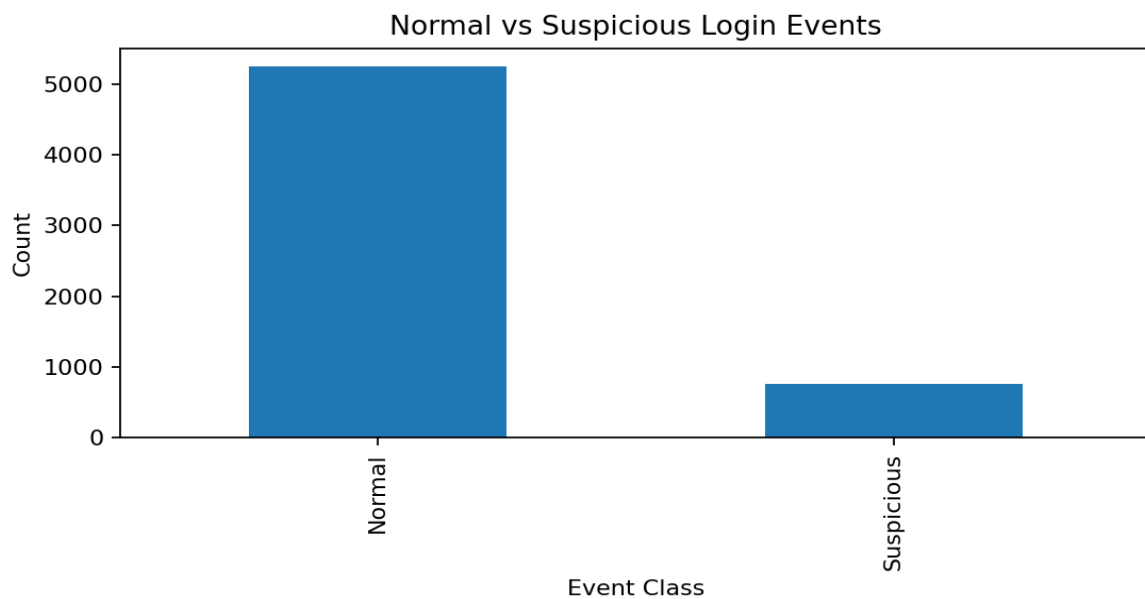


Figure 1. Normal versus suspicious login-event distribution.

Class distribution is important because a classifier can obtain high accuracy by favoring the majority class. For that reason, this project reports precision, recall and F1-score as well.

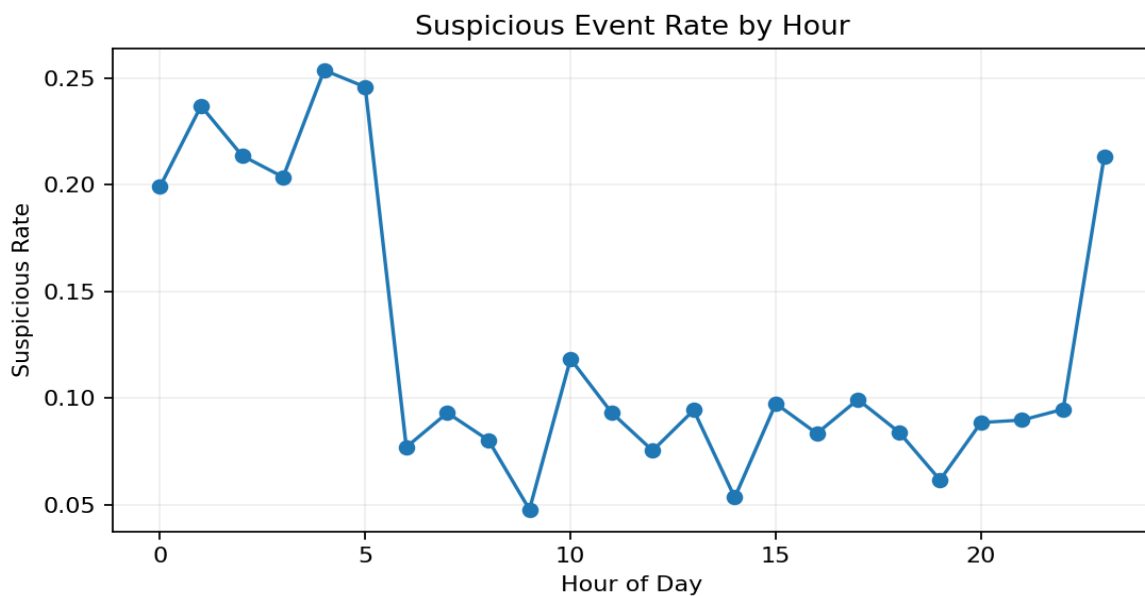


Figure 2. Suspicious-event rate by hour of day.

The hourly plot provides an exploratory view of the relationship between access time and suspicious-event frequency in the synthetic dataset. It is not intended as a universal rule for real users.

5. Security Indicators and Preprocessing

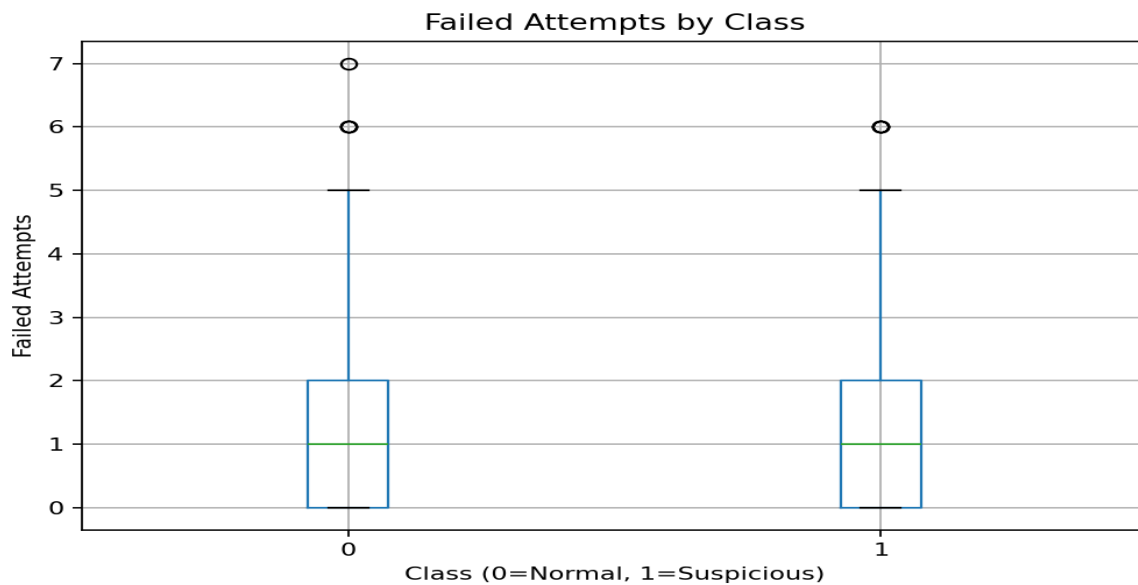


Figure 3. Failed-attempt distribution by target class.

Repeated failed authentication attempts may be useful as a signal. However, legitimate users can also create failed attempts because of forgotten passwords, keyboard errors, password-manager issues, or configuration problems.

Preprocessing pipeline

Numerical missing values are replaced using the median of the training feature. Categorical missing values are replaced using the most frequent category. Categorical features are one-hot encoded and numerical features are standardized.

Transformations are fitted on training data and then applied to test data, reducing the risk of test-set information leaking into training.

Train/test split: 80 percent training and 20 percent testing with stratification and random state 42.

Class imbalance: the training set is balanced by upsampling the minority class. The test set remains untouched so evaluation retains the original distribution.

6. Machine Learning Threat Classification

Logistic Regression was selected as the baseline because it is comparatively simple, fast, and interpretable. It produces a positive-class score that can be converted into a suspicious/not-suspicious decision using a threshold.

Training workflow

1. Load the CSV dataset.
2. Separate predictors and target.
3. Split into training and test sets.
4. Balance the training set only.
5. Fit preprocessing.
6. Train Logistic Regression.
7. Predict on the untouched test set.
8. Calculate evaluation metrics.
9. Inspect the confusion matrix.

Metrics

Accuracy is the proportion of correct predictions. Precision measures the reliability of suspicious flags. Recall measures coverage of suspicious events. F1-score combines precision and recall.

Metric	Logistic Regression
Accuracy	0.6500
Precision	0.2070
Recall	0.6291
F1-score	0.3115
Training time	0.0712 s

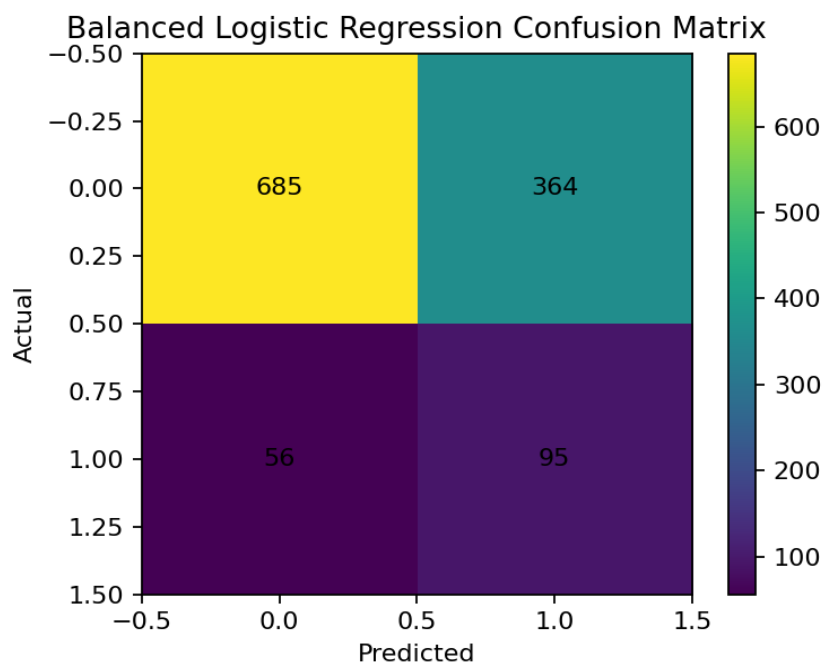


Figure 4. Logistic Regression confusion matrix.

7. Deep Learning Prototype

The assignment requires a simple neural-network implementation using scikit-learn MLPClassifier or a basic Keras model and comparison with the ML model on the same dataset.

An MLPClassifier was selected because it provides a straightforward neural-network implementation. The network contains hidden layers with 32 and 16 neurons, ReLU activation, Adam optimization, and early stopping.

Architecture: Input features -> dense layer (32, ReLU) -> dense layer (16, ReLU) -> binary output.

A neural network can represent nonlinear relationships between features. In cyber security, interactions between access time, device changes, traffic volume, reputation and authentication failures may be more complex than a single linear boundary. However, added complexity does not automatically mean better operational performance.

Metric	MLP Neural Network
Accuracy	0.7142
Precision	0.2363
Recall	0.5695
F1-score	0.3340
Training time	0.8498 s

The MLP result should be interpreted as an experiment on a synthetic tabular dataset, not as proof of general superiority over simpler models.

8. ML vs Neural Network Comparison

Metric	Logistic Regression	MLP	Meaning
Accuracy	0.650	0.714	Overall correctness
Precision	0.207	0.236	Reliability of flags
Recall	0.629	0.570	Suspicious-event coverage
F1-score	0.311	0.334	Precision/recall balance
Training time	0.071s	0.850s	Computational cost

Interpretation

The models demonstrate different precision and recall trade-offs on the same feature representation. Logistic Regression is comparatively easy to inspect and provides a useful baseline. The MLP can model nonlinear relationships but introduces additional complexity and tuning considerations.

Accuracy alone is insufficient for security classification. A model can appear accurate while missing suspicious events, particularly when the suspicious class is relatively small. Precision, recall, F1-score, confusion matrices and operational costs should be considered together.

Practical feasibility: both approaches are lightweight enough for this educational prototype. A production choice would require representative data, time-based validation, operational cost analysis, explainability requirements and monitoring.

Limitation: this comparison uses one synthetic dataset and one fixed train/test split and should not be generalized to real enterprise environments.

9. Ethical Reflection and AI Challenges

The assignment asks students to simulate false positives, consider biased data, modify thresholds or model parameters, and document ethical risks, privacy concerns and explainability limitations.

False positives. A legitimate event may be classified as suspicious. In a real authentication system this could lock accounts, interrupt employees, increase support workload, or reduce trust in security controls.

False negatives. Suspicious activity may be classified as normal. Depending on the application, this could allow an attacker to continue activity without escalation.

Bias. Security data can contain historical or sampling bias. Overrepresentation of one location, device family, or user population can create correlations that do not generalize. Error rates should be monitored across relevant populations where appropriate.

Privacy. Real security logs can contain sensitive information. Organizations should minimize collection, restrict access, define retention periods, protect stored data, and avoid unnecessary personal identifiers. This project uses synthetic data to reduce that risk.

Explainability. Analysts should understand why an event was flagged. Logistic Regression is relatively easier to inspect; neural networks can be harder to interpret. Explainability can help analysts investigate mistakes but does not guarantee correctness.

Human oversight. A responsible architecture should allow analyst review, investigation of supporting evidence, appropriate overrides, audit trails, and continuous model monitoring.

10. Threshold Experiment and Operational Trade-offs

The Logistic Regression score was evaluated at thresholds of 0.3, 0.5 and 0.7. The purpose is to demonstrate how changing the threshold affects false positives and false negatives rather than identifying a universally correct production threshold.

Threshold	Accuracy	Precision	Recall	False Positives	False Negatives
0.3	0.313	0.144	0.901	809	15
0.5	0.650	0.207	0.629	364	56
0.7	0.853	0.397	0.318	73	103

Lower threshold: more events are flagged, potentially increasing recall while also increasing legitimate events that require investigation.

Middle threshold: provides a conventional reference point for the experiment, but should not be assumed to be optimal for production.

Higher threshold: requires stronger model confidence and may reduce false alarms while increasing missed suspicious events.

Real threshold selection should consider the relative costs of account disruption, analyst workload, missed attacks, incident severity, user experience, and available secondary verification.

Operational principle: the model should be treated as one signal within a broader risk-scoring and analyst-review process rather than as the sole basis for automatically blocking an account.

11. Limitations and Future Improvements

Synthetic data limitation: the dataset is designed for education and reproducibility and cannot represent the full diversity of enterprise authentication systems or attacker behavior.

Limited features: real systems may combine authentication logs with endpoint telemetry, DNS events, process behavior, geolocation history, device fingerprints, network flows and threat-intelligence feeds.

Static validation: cyber threats evolve. A random train/test split does not test concept drift or future performance. A stronger study would use time-based validation and an external holdout dataset.

Future models: Decision Trees, Random Forest, Gradient Boosting, XGBoost, calibrated classifiers, and anomaly-detection methods could be compared. Cross-validation and hyperparameter tuning could also be introduced.

Explainability and monitoring: future versions could provide feature contributions for alerts, analyst feedback, drift detection, performance dashboards, and model-version monitoring.

Security of AI: production AI systems can themselves be attacked through adversarial inputs, data poisoning, model theft, and unauthorized access. These are outside the current Unit-1 scope.

Deployment: a future version could expose the classifier through a REST API or dashboard and integrate it with a simulated SIEM pipeline, with audit logs and safeguards for automated response.

12. Conclusion, References and Submission Checklist

This mini-project implements an end-to-end AI-driven cyber security awareness and detection prototype aligned with the Unit-1 requirements. It covers security-event data, EDA, preprocessing, supervised ML, a neural-network prototype, quantitative evaluation, threshold analysis, and ethical reflection.

The results demonstrate why security AI should be evaluated as a complete decision-support workflow rather than a single accuracy number. Class imbalance makes accuracy potentially misleading, while precision and recall expose different operational risks. Threshold changes further demonstrate that catching more suspicious events and reducing false alarms can conflict.

The neural-network experiment introduces nonlinear classification, while Logistic Regression provides a simpler baseline. Neither should be considered production-ready from this experiment alone because the data is synthetic, the feature set is limited, and validation is controlled.

Ethical considerations are central: privacy, bias, explainability, false positives, false negatives, and human oversight should be considered alongside technical performance.

References and support resources

1. AI in Cyber Security Lab, ENSP355, Lab Assignment 01 - Foundations of AI in Cyber Security.
2. Scikit-learn documentation - preprocessing, Logistic Regression, MLPClassifier, train/test splitting and evaluation metrics.
3. NIST Cybersecurity Framework - listed as a support resource in the assignment.

Submission checklist

1. Student name and roll number included
2. Dataset
3. EDA plots
4. ML classifier
5. Neural-network prototype
6. Accuracy, precision, recall and F1-score
7. Threshold and ethical analysis
8. README.md
9. requirements.txt
10. Python source code
11. Technical PDF report

Appendix - Implementation Summary

Core workflow

1. Load cyber_login_events.csv
2. Separate X and y
3. Identify numeric and categorical columns
4. Build ColumnTransformer
5. Impute missing values
6. Standardize numerical variables
7. One-hot encode categorical variables
8. Stratified 80/20 split
9. Upsample minority class in training data
10. Train Logistic Regression
11. Train MLPClassifier
12. Evaluate predictions
13. Generate confusion matrix
14. Run threshold experiment

The implementation follows the assignment progression from workspace setup and threat analysis through machine learning, neural-network comparison, and ethical analysis.

Academic integrity note

The assignment states that external datasets and references must be cited and that copying code or reports without citation can result in zero marks. This project therefore identifies the dataset as synthetic and lists the assignment and supporting technical resources.

End of Report